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### Change detection device and related methods

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#### Abstract

A change detection device may include a variational autoencoder (VAE) configured to encode image data to generate a latent vector, and decode the latent vector to generate new image data. The change detection device may further include a controller configured to select a deep learning model from different deep learning models based upon the new image data and a game theory reward matrix, and process the new image data using the selected deep learning model to detect changes therein.

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**Inventors:** Rahmes; Mark D. (Melbourne, FL), Boudreaux; Glenn (Melbourne, FL), Watkins; William W. (Melbourne Beach, FL)

**Applicant:** EAGLE TECHNOLOGY, LLC (Melbourne, FL)

**Family ID:** 1000008821399

**Assignee:** EAGLE TECHNOLOGY, LLC (Melbourne, FL)

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*Primary Examiner:* Shah; Utpal D

*Attorney, Agent or Firm:* ALLEN, DYER, DOPPELT, + GILCHRIST, P.A. Attorneys at Law

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## **Background/Summary**

### TECHNICAL FIELD

(1) The present disclosure relates to the field of image processing, and more particularly, change detection in images such as geospatial image data and related methods.

### BACKGROUND

(2) Topographical models of geographical areas may be used for many applications. For example, topographical models may be used in flight simulators and for geographical planning activities. Furthermore, topographical models of man-made structures (e.g., cities) may be helpful in applications such as cellular antenna placement, urban planning, disaster preparedness and analysis, and mapping, for example.

(3) Various types of topographical models are presently being used. One common topographical model is the digital elevation model (DEM). A DEM is a sampled matrix representation of a geographical area which may be generated in an automated fashion by a computer. In a DEM, coordinate points are made to correspond with a height or elevation value. Different types of DEMs include digital terrain models (DTMs) which represent the bare earth without any objects (e.g., vegetation and buildings), and digital surface models (DSMs) which represent the bare earth along with the objects (e.g., vegetation and buildings) thereon.

(4) One particularly useful application of such models is image change detection. One example approach for change detection is set forth in U.S. Pat. No. 11,238,307 to Delay et al., which is assigned to the present Applicant, discloses a system including a memory and a processor cooperating therewith to obtain geospatial image data from a plurality of different types of sensors and generate a three-dimensional (3D) geospatial model therefrom. The processor may further determine a reference image within the 3D geospatial model based upon synthetically positioning an image sensor within the 3D geospatial model, and perform change detection between a collected image and the reference image based upon semantic change detection using deep learning.

(5) Despite the advantages of such systems, further developments in image change detection may be desirable in certain applications.

### SUMMARY

(6) A change detection device may include a variational autoencoder (VAE) configured to encode image data to generate a latent vector, and decode the latent vector to generate new image data. The change detection device may further include a controller configured to select a deep learning model from a plurality of different deep learning models based upon the new image data and a game theory reward matrix, and process the new image data using the selected deep learning model to detect changes therein.

- (7) In an example embodiment, the controller may include an analyzer configured to determine object and voxel data from the new image data, and a classifier configured to register the new image data to historical image data and determine changes thereto based upon the determined object and voxel data and the selected deep learning model. By way of example, the plurality of different deep learning models may include at least one of Adaptive Moment Estimation (ADAM), Stochastic Gradient Descent with Momentum (SGDM), and RMSProp deep learning models.
- (8) In one example implementation, the VAE may include an encoder configured to generate a mean vector and a standard deviation vector from the image data, and generate the latent vector from the mean vector and the standard deviation vector. In accordance with one example, the game theory reward matrix may be based upon a Kullback-Leibler (KL) divergence loss associated with the plurality of different deep learning model.
- (9) In an example embodiment, the controller may be configured to solve the game theory reward matrix using a linear program. By way of example, the image data may comprise geospatial image data.
- (10) A related method for change detection may include encoding image data using a VAE to generate a latent vector, and decoding the latent vector using the VAE to generate new image data. The method may further include selecting a deep learning model from a plurality of different deep learning models using a controller based upon the new image data and a game theory reward matrix, and processing the new image data using the controller with the selected deep learning model to detect changes therein.
- (11) A related non-transitory computer-readable medium is for a change detection device including a VAE and having computer-executable instructions for causing a controller of the change detection device to perform steps including causing the VAE to encode image data to generate a latent vector and decode the latent vector to generate new image data, and selecting a deep learning model from a plurality of different deep learning models based upon the new image data and a game theory reward matrix. A further step may include processing the new image data using the selected deep learning model to detect changes therein.
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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic block diagram of a computing device according to an example embodiment.
- (2) FIG. 2 is a schematic block diagram of an example implementation of the change detection device of FIG. 1.
- (3) FIG. 3 is an example Sobel edge gradient module which may be used with the change detection device of FIG. 2.
- (4) FIGS. 4A and 4B are before and after graphs, respectively, showing changes in image data by the device of FIG. 3.
- (5) FIGS. 5A and 5B are digital elevation model (DEM) views illustrating the use of a change mask to highlight areas of changes in DEM image data by the change detection device of FIG. 1.
- (6) FIGS. 6A and 6B are another set of DEM views illustrating the use of a change mask to highlight areas of changes in DEM image data by the change detection device of FIG. 1.
- (7) FIGS. 7A and 7B are histograms illustrating test samples in which no change was detected, and where change was detected, respectively, in accordance with an example embodiment.
- (8) FIG. 8 is a Receiver Operator Characteristic (ROC) curve illustrating expected results for change analysis on image data from different solvers in an example embodiment.
- (9) FIG. 9 is a flow diagram illustrating example method aspects associated with the change detection device of FIG. 1.

#### DETAILED DESCRIPTION

(10) Referring initially to FIG. 1, a change detection device **30** is now described which may be used for processing image data **33** with a selected deep learning model to detect changes therein. The change detection device **30** illustratively includes a variational autoencoder (VAE) **31** configured to encode image data to generate a latent vector, and decode the latent vector to generate new image data. The change detection device **30** further illustratively includes a controller **32** configured to select a deep learning model from among a plurality of different deep learning models based upon the new image data and a game theory reward matrix, and process the new image data using the selected deep learning model to detect changes therein and optionally identify or output changed images **34**. By way of example, the VAE **31** and controller **32** may be implemented using appropriate hardware (e.g., microprocessor, etc.) and non-transitory computer-readable medium having computer-executable instructions for performing the various operations described further below. The change detection device **30** may be implemented as a stand-alone computing device, or it may be distributed among one or more machines, such as in a cloud-computing configuration in some embodiments.

(11) Generally speaking, the change detection device **30** optimizes latent space using VAEs to help change-detection processing. Pixel-level changes may be detected during training of deep learning networks. The latent space is optimized using game-theoretic methods to choose the best stochastic gradient descent solver for each pixel, and simulated data may also be generated with the VAE output. In some embodiments, the approach set forth herein may be implemented with an intellimatics knowledgebase to provide a unique engine for producing orthorectified, solar shadow/reflectance corrected, time-filtered input to a fully automated change-detection process, for example.

(12) By way of background, current change detection algorithms vary, but many use an entropy calculation to measure differences. Such change detection algorithms may benefit from using deep learning and game theoretic optimization. Furthermore, VAEs differ from regular autoencoders in that they do not use the encoding-decoding process to reconstruct an input. Instead, they impose a probability distribution on the latent space, and learn the distribution so that the distribution of outputs from the decoder matches that of the observed data. Then, they sample from this distribution to generate new data.

(13) Referring additionally to FIGS. 2 and 3, an example implementation of the change detection device **30** is now described. The VAE **31** illustratively includes an encoder **35** which learns to compress (reduce) a batch of input images **34** into an encoded representation of a normal distribution in latent space provided by a neural network **36** (e.g., a convolutional neural network, CNN). A decoder **37** learns to reconstruct the original data from the encoded representation to be as close to the original input as possible. The latent space is the layer that contains the compressed representation of the input data.

(14) The VAE **31** differs from regular autoencoders in that it does not use the encoding-decoding process simply to reconstruct an input. Instead, the VAE **31** imposes a probability distribution on the latent space and learns the distribution so that the distribution of the outputs from the decoder **37** matches that of the observed data. The VAE **31** assumes that the source data has some sort of underlying probability distribution (such as Gaussian) and then attempts to find the parameters of the distribution.

(15) The change detection device **30** advantageously provides an effective way to generate synthetic data for training machine learning (ML) applications, such as image change detection. In particular, this may be done while maintaining the underlying statistical properties of the original dataset, it may be applicable to sensitive datasets where traditional data masking falls short of protecting the data, and it may provide faster methods of generating synthetic training data for ML applications.

(16) By way of background, a VAE is a generative system and serves a similar purpose as a

generative adversarial network. One main use of a VAE is to generate new data that is related to the original source data by sampling from the learned distribution. Utilizing the learned distribution provides a way of generating synthetic data that is reflective of naturally occurring variations, rather than simply replicating existing data samples. This new synthetic data may be utilized for additional training and testing analysis. Moreover, a VAE may be considered a generative model which may randomly generate new samples based on the learned distribution. However, unlike traditional generative models that require strong assumptions regarding data structures and long inference times, a VAE makes weak assumptions of the data which also leads to faster training.

(17) The VAE **31** forces input images onto an n-dimensional probability distribution (e.g., a 20-dimensional Gaussian in the present example), learns the associated parameters (e.g., the means and variances for a Gaussian distribution), and describes the data seen on image pixels with the resulting distribution. Synthetic data samples may be randomly generated from a probability distribution in latent space once the associated parameter value vectors are calculated.

(18) The change detection device **30** may utilize a two-step process to generate synthetic data samples by (1) using the VAE **31** to learn the statistical properties of the original dataset(s) sampled from the Operational Design Domain (ODD); and (2) using the controller **32** as an optimizer for sampling the learned distribution and applying algorithmic transformations (e.g., rotations, reflections and attenuation) that enable building of richer datasets to support the ML model V&V process. More particularly, this approach provides an enhanced VAE-based process flow to learn the distribution and associated statistical properties of the original dataset (ideally the distribution of data in the ODD). Input data is provided, which in the present example includes geospatial images **33**.

(19) The image data **33** can represent any aspect or aspects of one or more devices and/or processes of a distributed system of interest. In the example of geospatial imagery, the data may take the form of a voxel representation including a descriptor stack of parameters such as pixel intensity, collection parameters, visibility, occlusion, solar angle, time existence, persistence, etc.

(20) The VAE **31** further illustratively includes an optional image gradient Sobel edge detector **39** (FIG. **3**), which is used for pre-processing. This helps the deep learning CNN models to learn more quickly and with more accuracy. The encoder **35** forces the input data (images **33**) onto the multidimensional probability distribution. In accordance with one example implementation, a 20-dimensional Gaussian distribution may be used, although other distributions and dimensions may be utilized in different embodiments. The VAE **31** learns the means and variances of the data, and the resulting distribution describes the data.

(21) The encoder **35** generates a compressed representation of the input data utilizing various weights and biases. Weights are the parameters within the neural network **36** that transform input data within the network's hidden layers. Generally speaking, the neural network **36** is made up of a series of nodes. Within each node is a set of inputs, weight, and a bias value. As an input enters the node, it gets multiplied by a weight value, and the resulting output is either observed or passed to the next layer in the neural network **36**. The weights of the neural network **36** may be included within the hidden layers of the network. Within the neural network **36**, an input layer may take the input signals and pass them to the next layer. Next, the neural network **36** includes a series of hidden layers which apply transformations to the input data. It is within the nodes of the hidden layers that the weights are applied. For example, a single node may take the input data and multiply it by an assigned weight value, then add a bias before passing the data to the next layer. The final layer of the neural network **36** is known as the output layer. The output layer often tunes the inputs from the hidden layers to produce the desired numbers in a specified range.

(22) Weights and bias values are both learnable parameters inside the neural network **36**. The neural network **36** may randomize both the weight and bias values before learning initially begins. As training continues, both parameters may be adjusted toward the desired values and the correct output. The two parameters differ in the extent of their influence upon the input data. At its

simplest, bias represents how far off the predictions are from their intended value. Biases make up the difference between the function's output and its intended output. A low bias suggests that the neural network **36** is making more assumptions about the form of the output, whereas a high bias value makes less assumptions about the form of the output. Weights, on the other hand, can be thought of as the strength of the connection. Weight affects the amount of influence a change in the input will have upon the output. A low weight value will have no change on the input, and alternatively a larger weight value will more significantly change the output.

(23) The compressed representation of the input data is called the hidden vector. The mean and variance from the hidden vector are sampled and learned by the neural network **36**. Principal component analysis (PCA) of the hidden vector allows for the visualization of n-dimensional point clusters, e.g., 3-D point clusters, in the latent space. To make calculations more numerically stable, the range of possible values may be increased by making the network learn from the logarithm of the variances. Two vectors may be defined: one for the means, and one for the logarithm of the variances. Then, these two vectors may be used to create the distribution from which to sample.

(24) More particularly, the neural network **36** takes mean and the variance encodings generated after passing the test images through the encoder network **35** and performs PCA on the matrix containing the encodings for each of the images. Furthermore, the neural network **36** visualizes the latent space defined by the means and the variances in the three dimensions characterized by the three first principal components, and initializes new encodings sampled from a normal distribution and outputs the images generated when these encodings pass through the decoder network **37**.

(25) The decoder **37** generates synthetic output data. The controller **32** functions as an optimizer which uses an ensemble of solvers with a game theoretic implementation to create an output image with least image reconstruction error. In the illustrated example, the controller **32** includes an analyzer **46** configured to determine object and voxel data from the new image data, and a classifier **38** configured to register the new image data to historical image data and determine changes thereto based upon the determined object and voxel data and the selected deep learning model.

(26) More particularly, the controller **32** computes a gradient of loss function from the synthetic output data, and picks the best update based upon the ensemble of solvers provided by the classifier **38**. More particularly, the optimization process is iterated via reparameterization to handle sampling of the hidden vector during backpropagation (an algorithm for training neural networks). In the illustrated example, an ensemble of models is generated using the three different solvers, namely an Adaptive Moment Estimation (ADAM) solver, a Stochastic Gradient Descent with Momentum (SGDM) solver, and a Root Mean Squared Propagation (RMSProp) solver, although different solvers may be used in different embodiments. The values from the loss function (evidence lower bound or ELBO, reconstruction, and Kullback-Leibler or KL loss) may be used in a game theoretic implementation to determine the optimal model to use per test sample. The loss is used to compute the gradients of the solvers.

(27) In an example implementation, the VAE **31** may implement a loss step, in which it passes the encoding generated by the sampling step through the decoder network **37** and determines the loss, which is then used to compute the gradients. The loss in VAEs, also called the evidence lower bound (ELBO) loss, is defined as a sum of two separate loss terms: reconstruction loss+Kullback-Leibler (KL) loss or divergence. More particularly, reconstruction loss measures how close the decoder output is to the original input by using the mean-squared error (MSE). KL divergence measures the difference between two probability distributions. Minimizing the KL loss in this case means ensuring that the learned means and variances are as close as possible to those of the target (normal) distribution. The practical effect of including the KL loss term is to pack clusters learned due to reconstruction loss tightly around the center of the latent space, forming a continuous space from which to sample.

(28) Example code for the encoder network **35** may be as follows:

(29) TABLE-US-00001 encoderLG = layerGraph([ x  
 imageInputLayer(imageSize, 'Name', 'input\_encoder', 'Normalization', 'none') •  
 convolution2dLayer(3,4,'Padding','same','Name','conv\_1') x  
 batchNormalizationLayer('Name','BN\_1') x reluLayer('Name','relu\_1') •  
 maxPooling2dLayer(1,'Stride',1, 'Name','max1') x  
 convolution2dLayer(3,8,'Padding','same','Stride',2, 'Name','conv\_2') •  
 batchNormalizationLayer('Name','BN\_2') x reluLayer('Name','relu\_2') x  
 maxPooling2dLayer(1,'Stride', 1, 'Name','max2') •  
 convolution2dLayer(3,16,'Padding','same','Stride',2, 'Name','conv\_3') x  
 batchNormalizationLayer('Name','BN\_3') • reluLayer('Name','relu\_3') •  
 maxPooling2dLayer(1,'Stride', 1, 'Name','max3') •  
 convolution2dLayer(3,32,'Padding','same','Stride',2, 'Name','conv\_4') •  
 batchNormalizationLayer('Name','BN\_4') • reluLayer('Name','relu\_4') x  
 maxPooling2dLayer(1,'Stride', 1, 'Name','max4') •  
 convolution2dLayer(3,64,'Padding','same','Stride',2, 'Name','conv\_5') x  
 batchNormalizationLayer('Name','BN\_5') • reluLayer('Name','relu\_5') •  
 maxPooling2dLayer(1,'Stride', 1, 'Name','max5') x  
 convolution2dLayer(3,128,'Padding','same','Stride',2, 'Name','conv\_6') •  
 batchNormalizationLayer('Name','BN\_6') x reluLayer('Name','relu\_6') •  
 fullyConnectedLayer(2.sup.xlatentDim, 'Name', 'fc')]);

Furthermore, code for the decoder network **37** may be as follows:

(30) TABLE-US-00002 decoderLG = layerGraph([1 1 latentDim], 'Name'  
 'i', 'Normalization', 'none') transposedConv2dLayer(3, 64, 'Cropping', 'same', 'Stride', 2, 'Name',  
 'transpose1') reluLayer('Name','relu1') transposedConv2dLayer(3, 32, 'Cropping', 'same',  
 'Stride', 2, 'Name', 'transpose2') reluLayer('Name','relu2') transposedConv2dLayer(3, 16,  
 'Cropping', 'same', 'Stride', 2, 'Name', 'transpose3') reluLayer('Name','relu3')  
 transposedConv2dLayer(3, 8, 'Cropping', 'same', 'Stride', 2, 'Name', 'transpose4')  
 reluLayer('Name','relu4') transposedConv2dLayer(3, 4, 'Cropping', 'same', 'Stride', 2, 'Name',  
 'transpose5') reluLayer('Name','relu5') transposedConv2dLayer(3, 2, 'Cropping', 'same', 'Stride',  
 2, 'Name', 'transpose6') reluLayer('Name','relu6') transposedConv2dLayer(3, 1, 'Cropping',  
 'same', 'Stride', 2, 'Name', 'transpose7') ]);

(31) The controller **32** creates a reward matrix, with pixel values and different solvers. The reward matrix may be an M×C matrix, where M is the number of models in the ensemble and C is the number of classes. In the example implementation, one model is used for each solver, for a total of three models, namely ADAM, SGDM, and RMSProp. The controller **32** solves the matrix for each pixel and the reward matrix uses reconstruction and KL loss scores, or responses based on number of pixel values to determine a goodness of fit. Scores in a linear program may be used to optimally choose which deep learning model to use per pixel. The matrix is constructed and solved with a linear program such as an interior-point algorithm, e.g., the primal-dual method, which may be feasible for convergence. A primal standard form may be used to calculate optimal tasks and characteristics as follows:

maximize  $f(x)$  s.t.

$Ax \leq b$

$x \geq 0$

(32) An example of latent space change detection is illustrated in the 3D plots **40** and **43** of FIGS. **4A** and **4B** at first and second times, respectively. At the first time (plot **40**), a certain number of data points **41** at the bottom of the plot correspond to changed pixels, while the point **42** at the top of the plot represent unchanged pixels. At the second time (plot **43**), more changes have occurred, resulting in less unchanged plot points **42** and more changed plot points **41**.

(33) Examples of change detection are shown in FIGS. **5A**, **5B** and **6A**, **6B**. In both of these

examples, the geospatial image **33** is provided as an input to the device **30**, and the output image **34** includes a change mask **45** overlayed on the image to show regions or points where change has been detected. It should be noted that a change mask need not be continuous, and may encompass several spaced-apart pixel regions within the image **34** as shown. It should be noted that a change mask **45** need not be provided in all embodiments, and that other suitable approaches for illustrating changes in the images **33** may be used as well.

(34) Referring additionally to FIGS. **7A** and **7B**, example histograms **71** and **72** showing unchanged/changed pixel test samples, respectively, are now described. A zTest may be used to determine if the new signal distribution belongs to any existing distributions. The controller **32** may loop through all distributions and keep the highest p value for each zTest. A high p value means that the new distribution is already in the training data. Then, the controller **32** subtracts 1 from these scores for **H0** and **H1**. More particularly, the graph **71** (**H0**) represents the values that do not belong to the new data distribution. Furthermore, the graph **72** (**H1**) represents the values that do belong to the new data distribution.

(35) Furthermore, a ROC curve **73** illustrated in FIG. **8** provides a plot of all threshold **H0** and **H1** values, in which the y-axis is the probability of detection (PD) and the x-axis is the probability of false alarm (PF). In the illustrated example, the plot **74** is for game theory, the plot **75** corresponds to SGDM, the plot **76** corresponds to ADAM, and the plot **77** corresponds to RMS Propagation. Of the three solvers, SGDM is the best performing model in this particular case and accordingly would be selected as the deep learning model for change detection.

(36) The above-described approach advantageously utilizes the VAE **31** for image change detection. This approach also allows for the use of image gradient and 2D convolution as pre-processing to input image patches, as well as an encoder-decoder architecture in deep learning convolutional neural network. Furthermore, use of game theoretic optimization helps select the best performing deep learning model with an ensemble of solvers (e.g., SGDM, ADAM, RMSProp) to find minimum cost function loss (error). The present approach may provide further advantages including: providing a 3D p-value statistical test to measure anomaly detection accuracy and may represent results with ROC curves; automatically selecting a 3D view of latent space clusters that shows most separation of test hypotheses; and calculation of the probability of the most likely latent space cluster to which new sample image might belong to the latent space distribution.

(37) Turning now to the flow diagram **90** of FIG. **9**, beginning at Block **91**, a related method for change detection illustratively includes encoding image data using a VAE to generate a latent vector, at Block **92**, and decoding the latent vector using the VAE to generate new image data, at Block **93**. The method further illustratively includes selecting a deep learning model from a plurality of different deep learning models using the controller **32** based upon the new image data and a game theory reward matrix, at Block **94**, and processing the new image data using the controller with the selected deep learning model to detect changes therein, at Block **95**, as discussed further above. The method of FIG. **9** illustratively concludes at Block **96**.

(38) A related non-transitory computer-readable medium is for a change detection device **30** including a VAE **31** and having computer-executable instructions for causing a controller of the change detection device to perform steps including causing the VAE to encode image data to generate a latent vector and decode the latent vector to generate new image data, and selecting a deep learning model from a plurality of different deep learning models based upon the new image data and a game theory reward matrix. A further step may include processing the new image data using the selected deep learning model to detect changes therein, as discussed further above.

(39) Many modifications and other embodiments will come to the mind of one skilled in the art having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is understood that the disclosure is not to be limited to the specific embodiments disclosed, and that modifications and embodiments are intended to be included within the scope of the appended claims.

## Claims

1. A change detection device comprising: a variational autoencoder (VAE) configured to encode image data to generate a latent vector, and decode the latent vector to generate new image data; and a controller configured to select a deep learning model from a plurality of different deep learning models based upon the new image data and a game theory reward matrix, and process the new image data using the selected deep learning model to detect changes therein.
2. The change detection device of claim 1 wherein the controller comprises: an analyzer configured to determine object and voxel data from the new image data; and a classifier configured to register the new image data to historical image data and determine changes thereto based upon the determined object and voxel data and the selected deep learning model.
3. The change detection device of claim 1 wherein the plurality of different deep learning models comprises one of Adaptive Moment Estimation (ADAM), Stochastic Gradient Descent with Momentum (SGDM), and RMSProp deep learning models.
4. The change detection device of claim 1 wherein the VAE comprises an encoder configured to generate a mean vector and a standard deviation vector from the image data, and generate the latent vector from the mean vector and the standard deviation vector.
5. The change detection device of claim 1 wherein the game theory reward matrix is based upon a Kullback-Leibler (KL) divergence loss associated with the selected deep learning model.
6. The change detection device of claim 1 wherein the controller is configured to solve the game theory reward matrix using a linear program.
7. The change detection device of claim 1 wherein the image data comprises geospatial image data.
8. A method for change detection comprising: encoding image data using a variational autoencoder (VAE) to generate a latent vector; decoding the latent vector using the VAE to generate new image data; selecting a deep learning model from a plurality of different deep learning models using a controller based upon the new image data and a game theory reward matrix; and processing the new image data using the controller with the selected deep learning model to detect changes therein.
9. The method of claim 8 wherein the controller comprises an analyzer and a classifier, and further comprising: determining object and voxel data from the new image data using the analyzer; and registering the new image data to historical image data and determining changes thereto based upon the determined object and voxel data and the selected deep learning model using the classifier.
10. The method of claim 8 wherein the plurality of different deep learning models comprises one of Adaptive Moment Estimation (ADAM), Stochastic Gradient Descent with Momentum (SGDM), and RMSProp deep learning models.
11. The method of claim 8 wherein encoding comprises generating a mean vector and a standard deviation vector from the image data, and generating the latent vector from the mean vector and the standard deviation vector.
12. The method of claim 8 wherein the game theory reward matrix is based upon a Kullback-Leibler (KL) divergence loss associated with the selected deep learning models.
13. The method of claim 8 wherein selecting comprises solving the game theory reward matrix using a linear program.
14. The method of claim 8 wherein the image data comprises geospatial image data.
15. A non-transitory computer-readable medium for a change detection device including a variational autoencoder (VAE) and having computer-executable instructions for causing a controller of the change detection device to perform steps comprising: causing the VAE to encode image data to generate a latent vector, and decode the latent vector to generate new image data; selecting a deep learning model from a plurality of different deep learning models based upon the new image data and a game theory reward matrix; and processing the new image data using the

selected deep learning model to detect changes therein.

16. The non-transitory computer readable medium of claim 15 further having computer-executable instructions for causing the controller of the change detection device to perform steps comprising: determining object and voxel data from the new image data; and registering the new image data to historical image data and determine changes thereto based upon the determined object and voxel data and the selected deep learning model.

17. The non-transitory computer readable medium of claim 15 wherein the plurality of different deep learning models comprises one of Adaptive Moment Estimation (ADAM), Stochastic Gradient Descent with Momentum (SGDM), and RMSProp deep learning models.

18. The non-transitory computer readable medium of claim 15 further having computer-executable instructions for causing the controller of the change detection device to cause the VAE to generate a mean vector and a standard deviation vector from the image data, and generate the latent vector from the mean vector and the standard deviation vector.

19. The non-transitory computer readable medium of claim 15 wherein the game theory reward matrix is based upon a Kullback-Leibler (KL) divergence loss associated with the selected deep learning model.

20. The non-transitory computer readable medium of claim 15 further having computer-executable instructions for causing the controller of the change detection device to solve the game theory reward matrix using a linear program.

21. The non-transitory computer readable medium of claim 15 wherein the image data comprises geospatial image data.

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